

Representation Padding in Algorithm Class 1: A Literal Negative Answer and the Remaining Full-Row-Rank Question

Abstract

Liu, Lin, and Xu ask whether their lower bound

$$\Omega\left(\kappa\left(\begin{smallmatrix} \bar{A} \\ A \end{smallmatrix}\right) L_f \Delta F_0 \varepsilon^{-2}\right)$$

for Algorithm Class 1 can be attained up to logarithmic factors on their Problem Class 1. We prove that the question, read literally and without a restriction on the chosen representation $g(x) = \bar{g}(\bar{A}x + \bar{b})$, has a negative answer. Any valid representation can be enlarged by coordinates that the outer function ignores. This leaves the optimization problem, the proximal oracle, and the exact Algorithm Class 1 span restrictions unchanged, while making the displayed condition number arbitrarily close to one. Applying this operation to the authors' own hard family gives represented instances with a globally 1-Lipschitz outer function, condition number at most $\sqrt{2}$, and uniformly bounded $L_f \Delta F_0 \varepsilon^{-2}$, yet requiring a number of oracle calls that grows linearly with the dimension parameter. Thus no uniform bound of the displayed order, even with standard polylogarithmic factors, can hold over unrestricted representations.

The result is a *solution of the literal open question*, but only *partial progress on the contextual conjecture* in the paper. Immediately before posing the question, the authors conjecture tightness under the additional requirement that $\begin{pmatrix} \bar{A} \\ A \end{pmatrix}$ have full row rank. Our padding necessarily violates that requirement. We prove, moreover, that the authors' hard family cannot be re-represented with full row rank and a substantially smaller condition number: an intrinsic principal-angle obstruction forces every such representation to retain condition number of the same order.

Status of the result. The unrestricted formulation in Section 4 of Liu–Lin–Xu is answered negatively. The full-row-rank formulation suggested by the preceding discussion is not solved here; the results below explain precisely why the representation-padding counterexample does not enter that regime.

1 The question and the exact oracle model

Consider

$$\min_{x \in \mathbb{R}^d} F_0(x) := f_0(x) + g(x) \quad \text{subject to} \quad Ax + b = 0, \tag{1}$$

where f_0 has L_f -Lipschitz gradient and g is proper, lower semicontinuous, and convex. Problem Class 1 of Liu, Lin, and Xu [1] consists of represented instances for which

$$g(x) = \bar{g}(\bar{A}x + \bar{b}), \tag{2}$$

with $\bar{g}$ proper, lower semicontinuous, and convex, $\text{relint}(\text{dom } \bar{g}) \neq \emptyset$, and $\text{dom } \bar{g}$ not a singleton; the class also requires $\inf_x F_0(x) > -\infty$. A point x is ε -stationary if there exists γ such that

$$\max \left\{ \text{dist}(0, \nabla f_0(x) + A^\top \gamma + \partial g(x)), \|Ax + b\| \right\} \leq \varepsilon. \quad (3)$$

For a nonzero matrix M , write

$$\kappa(M) := \frac{\sigma_{\max}(M)}{\sigma_{\min}^+(M)} = \left(\frac{\lambda_{\max}(MM^\top)}{\lambda_{\min}^+(MM^\top)} \right)^{1/2}. \quad (4)$$

The first oracle in [1] returns

$$(\nabla f_0(x), Ax, A^\top z, \text{prox}_{\eta g}(x)).$$

The oracle list alone is not the definition of Algorithm Class 1. Its exact linear-span restriction is as follows: from $x^{(0)}$, for every $t \geq 1$,

$$x^{(t)} \in \text{span}\{\xi^{(t)}, \text{prox}_{\eta_t g}(\xi^{(t)})\}, \quad (5)$$

$$\xi^{(t)} \in \text{span}\left\{x^{(s)}, \nabla f_0(x^{(s)}), A^\top b, A^\top Ax^{(s)} : 0 \leq s < t\right\}, \quad \eta_t > 0. \quad (6)$$

In particular, neither $\bar{A}$, $\bar{b}$, nor $\bar{g}$ appears in the allowable spans.

Let

$$B := \begin{pmatrix} \bar{A} \\ A \end{pmatrix}, \quad Q := L_f \Delta F_0 \varepsilon^{-2}, \quad \Delta F_0 := F_0(x^{(0)}) - \inf_x F_0(x).$$

The literal question in Section 4 of [1] is whether an Algorithm Class 1 method can attain a uniform complexity

$$\tilde{O}(\kappa(B)Q). \quad (7)$$

The discussion immediately preceding that question is narrower: it conjectures near-tightness when $\bar{g}$ is Lipschitz and B has full row rank. We keep these two formulations separate.

2 Representation normalization and padding

The key point is that a representation in (2) is not intrinsic data for Algorithm Class 1.

Lemma 2.1 (Normalization and padding). *Suppose $g(x) = \bar{g}(\bar{A}x + \bar{b})$ is a valid Problem Class 1 representation. For arbitrary $\alpha, \rho > 0$, define*

$$\hat{A}_{\alpha, \rho} := \begin{pmatrix} \alpha \bar{A} \\ \rho I_d \end{pmatrix}, \quad \hat{b}_{\alpha, \rho} := \begin{pmatrix} \alpha \bar{b} \\ 0 \end{pmatrix}, \quad \hat{g}_{\alpha, \rho}(u, v) := \bar{g}(u/\alpha). \quad (8)$$

Then (8) is another valid representation of the same function g . If $\bar{g}$ is globally $L_{\bar{g}}$ -Lipschitz in the Euclidean norm and $\alpha \geq \max\{1, L_{\bar{g}}\}$, then $\hat{g}_{\alpha, \rho}$ is globally 1-Lipschitz.

Set

$$C_\alpha := \begin{pmatrix} \alpha \bar{A} \\ A \end{pmatrix}, \quad \hat{B}_{\alpha, \rho} := \begin{pmatrix} \alpha \bar{A} \\ \rho I_d \\ A \end{pmatrix}.$$

The matrix $\hat{B}_{\alpha, \rho}$ has full column rank and

$$\kappa(\hat{B}_{\alpha, \rho})^2 = \frac{\rho^2 + \lambda_{\max}(C_\alpha^\top C_\alpha)}{\rho^2 + \lambda_{\min}(C_\alpha^\top C_\alpha)}. \quad (9)$$

Consequently, if $\rho \geq \|C_\alpha\|$, then

$$\kappa(\widehat{B}_{\alpha,\rho}) \leq \sqrt{2}, \quad (10)$$

and $\kappa(\widehat{B}_{\alpha,\rho}) \rightarrow 1$ as $\rho \rightarrow \infty$.

Proof. For every x ,

$$\widehat{g}_{\alpha,\rho}(\widehat{A}_{\alpha,\rho}x + \widehat{b}_{\alpha,\rho}) = \bar{g}((\alpha\bar{A}x + \alpha\bar{b})/\alpha) = g(x).$$

The function $\widehat{g}_{\alpha,\rho}$ is proper, lower semicontinuous, and convex. Its domain is $\alpha \operatorname{dom}(\bar{g}) \times \mathbb{R}^d$, whose relative interior is $\alpha \operatorname{relint}(\operatorname{dom} \bar{g}) \times \mathbb{R}^d$; it is not a singleton. Hence it is a valid outer function. If $\bar{g}$ is $L_{\bar{g}}$ -Lipschitz and $\alpha \geq L_{\bar{g}}$, then

$$|\widehat{g}_{\alpha,\rho}(u, v) - \widehat{g}_{\alpha,\rho}(u', v')| \leq \frac{L_{\bar{g}}}{\alpha} \|u - u'\| \leq \|(u, v) - (u', v')\|.$$

Finally,

$$\widehat{B}_{\alpha,\rho}^\top \widehat{B}_{\alpha,\rho} = C_\alpha^\top C_\alpha + \rho^2 I_d.$$

Because $\rho > 0$, this matrix is positive definite, and its eigenvalues are exactly the positive squared singular values of $\widehat{B}_{\alpha,\rho}$. This proves (9). Since $\lambda_{\max}(C_\alpha^\top C_\alpha) = \|C_\alpha\|^2$ and $\lambda_{\min}(C_\alpha^\top C_\alpha) \geq 0$,

$$\kappa(\widehat{B}_{\alpha,\rho})^2 \leq 1 + \frac{\|C_\alpha\|^2}{\rho^2} \leq 2.$$

The limiting statement is immediate from (9). $\square$

Corollary 2.2 (The unrestricted represented condition number is degenerate). *For every underlying Problem Class 1 instance,*

$$\inf \left\{ \kappa \begin{pmatrix} \widetilde{\bar{A}} \\ \bar{A} \end{pmatrix} : g = \widetilde{\bar{g}}(\widetilde{\bar{A}} \cdot + \widetilde{\bar{b}}) \text{ is a valid representation} \right\} = 1.$$

Proof. Condition numbers are at least one, and Lemma 2.1 gives valid representations whose condition numbers converge to one. $\square$

Lemma 2.3 (Algorithm Class 1 is invariant under re-representation). *Fix $f_0, g, A, b, x^{(0)}$. If two triples $(\bar{g}, \bar{A}, \bar{b})$ represent the same function g , then they induce exactly the same first oracle and exactly the same collection of sequences satisfying (5)–(6). Hence every lower bound that holds for all Algorithm Class 1 methods for one representation holds for every other representation of the same g .*

Proof. The oracle and both span restrictions depend only on f_0, g, A, b and $x^{(0)}$. In particular, the proximal map is the same because the function g itself is the same. No term involving $\bar{g}$, $\bar{A}$, or $\bar{b}$ occurs in (5)–(6). $\square$

3 A hard sequence with bounded padded condition number

We now apply the preceding lemmas to the hard family constructed in [1, Definition 2 and the proof of Theorem 1]. Using that family rather than a separate smooth lower bound avoids any mismatch of oracle models or quantifiers.

Proposition 3.1 (A direct specialization of the Liu–Lin–Xu family). *Fix $\varepsilon_0 = 1/2$ and $L_f = 1$. For every integer $r \geq 1$, there is an instance $\mathcal{P}_r$ in Problem Class 1, initialized at $x^{(0)} = 0$, with the following properties.*

(i) *The construction parameters can be chosen as*

$$m_1 = 2, \quad m_2 = r, \quad m = 3m_1m_2 = 6r, \quad \bar{d} = 5.$$

Consequently,

$$d = 30r, \quad n = 15r, \quad \bar{n} = 15r - 5. \quad (11)$$

(ii) *With the authors' original representation, $B_r^0 = \begin{pmatrix} \bar{A}_r \\ A_r \end{pmatrix}$ is a row permutation of their matrix H_r , has full row rank, and satisfies*

$$\frac{m}{4} \leq \kappa(B_r^0) < m. \quad (12)$$

(iii) *The original outer function is finite-valued and globally Lipschitz.*

(iv) *The initial gap obeys*

$$Q_r := L_f \Delta F_{0,r} \varepsilon_0^{-2} \leq 15000\pi^2. \quad (13)$$

(v) *Every Algorithm Class 1 method requires at least*

$$2 + \frac{m(\bar{d} - 2)}{6} = 3r + 2 \quad (14)$$

first-oracle calls to produce an ε_0 -stationary point.

Moreover, the free coefficient β_r in the construction may be selected so that all matrix norms, Lipschitz constants, and scalar coefficients in the represented data are bounded by a polynomial in r .

Proof. The cited construction permits even $m_1 \geq 2$, arbitrary positive m_2 , and odd $\bar{d} \geq 5$. Substitution gives (11) from $d = m\bar{d}$, $n = (m - 3m_2)\bar{d}$, and $\bar{n} = (3m_2 - 1)\bar{d}$. The stacked matrix is obtained by partitioning the rows of $H_r = mL_f(J_m \otimes I_{\bar{d}})$; it is therefore a row permutation of H_r , and (12) is Lemma 3 of [1].

The outer function is

$$\bar{g}_r(y) = \frac{\beta_r}{mL_f} \|y\|_1,$$

so it is globally Lipschitz in the Euclidean norm. Lemma 2(a) of [1], together with $g_r(0) = 0$ and $g_r \geq 0$, gives

$$\Delta F_{0,r} \leq \frac{3000\pi^2 \bar{d} \varepsilon_0^2}{L_f}.$$

Taking $\bar{d} = 5$ yields (13).

The zero-chain argument in the proof of Theorem 1 of [1] proves that no Algorithm Class 1 iterate obtained in at most $1 + m(\bar{d} - 2)/6$ calls is ε_0 -stationary. Thus at least $2 + m(\bar{d} - 2)/6 = 3r + 2$ calls are required, which is (14).

For the final statement, choose, for example,

$$\beta_r = (50\pi + 2 + \|A_r\|)\sqrt{m}\varepsilon_0,$$

which satisfies the strict inequality required in the construction. Since $\|A_r\| \leq \|H_r\| < 2mL_f = O(r)$, one has $\beta_r = O(r^{3/2})$. Also $\bar{n} = O(r)$, so the Euclidean Lipschitz constant $\beta_r\sqrt{\bar{n}}/(mL_f)$ of $\bar{g}_r$ is $O(r)$. The remaining coefficients in the explicit formulas of [1] are likewise polynomially bounded in m when L_f and ε_0 are fixed. $\square$

Theorem 3.2 (Literal negative answer). *There exists a sequence of represented instances $\widehat{\mathcal{P}}_r$ in Problem Class 1 such that*

- (i) *the underlying optimization problem $(f_{0,r}, g_r, A_r, b_r)$ is exactly that of $\mathcal{P}_r$;*
- (ii) *the displayed outer function is globally 1-Lipschitz;*
- (iii) *with $\widehat{B}_r = \begin{pmatrix} \widehat{A}_r \\ A_r \end{pmatrix}$,*

$$\kappa(\widehat{B}_r) \leq \sqrt{2}; \quad (15)$$

- (iv) *$Q_r \leq 15000\pi^2$ and the total represented dimension is*

$$D_r := d + n + \widehat{n} = 90r - 5; \quad (16)$$

- (v) *every Algorithm Class 1 method still needs at least $3r + 2$ oracle calls to produce an ε_0 -stationary point.*

All standard numerical parameters of this sequence are polynomially bounded in r .

Proof. Let L_r be the Euclidean Lipschitz constant of the original $\bar{g}_r$, and set

$$\alpha_r := \max\{1, L_r\}, \quad C_r := \begin{pmatrix} \alpha_r \bar{A}_r \\ A_r \end{pmatrix}, \quad \rho_r := \max\{1, \|C_r\|\}.$$

Apply Lemma 2.1 with $(\alpha, \rho) = (\alpha_r, \rho_r)$. This gives

$$\widehat{A}_r = \begin{pmatrix} \alpha_r \bar{A}_r \\ \rho_r I_d \end{pmatrix}, \quad \widehat{b}_r = 0, \quad \widehat{g}_r(u, v) = \bar{g}_r(u/\alpha_r). \quad (17)$$

The represented function is still g_r , the outer function is 1-Lipschitz, and $\kappa(\widehat{B}_r) \leq \sqrt{2}$.

The new outer dimension is $\widehat{n} = \bar{n} + d = (15r - 5) + 30r = 45r - 5$, and hence $D_r = 30r + 15r + (45r - 5) = 90r - 5$. The objective gap, accuracy, and smoothness constant are unchanged, so (13) remains valid. Lemma 2.3 and Proposition 3.1(v) give the lower bound $3r + 2$.

Finally, Proposition 3.1 gives $L_r = O(r)$ and $\|\bar{A}_r\|, \|A_r\| = O(r)$. Thus $\alpha_r = O(r)$, $\|C_r\| = O(r^2)$, and $\rho_r = O(r^2)$. Together with the explicit source construction, this proves polynomial boundedness of the represented data. $\square$

Corollary 3.3 (No uniform polylogarithmic repair of the displayed bound). *There do not exist an Algorithm Class 1 method, constants $C > 0$ and $q \geq 0$, such that every represented instance in Problem Class 1 is solved within*

$$C(1 + \kappa(B)Q)(1 + \log(2 + d + n + \bar{n}))^q \quad (18)$$

first-oracle calls. In particular, the literal uniform guarantee (7) is false under the standard meaning of $\widetilde{O}$, where the suppressed factors are polylogarithmic in dimensions and polynomially bounded numerical parameters.

Proof. On $\widehat{\mathcal{P}}_r$, the bound (18) is $O((\log r)^q)$ by Theorem 3.2, whereas every Algorithm Class 1 method needs at least $3r + 2$ calls. This is impossible for large r . The same comparison applies to any fixed product of logarithms of the usual data parameters, because those parameters are polynomially bounded in r . $\square$

Remark 3.4 (What has been solved). *Corollary 3.3 is a negative solution to the unqualified question printed in Section 4 of [1]. It does not contradict the authors' lower bound: that lower bound is valid for the particular hard representation they construct. Rather, it shows that the same underlying Algorithm Class 1 instance can be assigned a different valid representation for which the displayed parameter no longer measures the difficulty.*

4 Why the full-row-rank conjecture is not resolved

The padded matrix in Theorem 3.2 has

$$n + \hat{n} = 15r + (45r - 5) = 60r - 5 > d = 30r,$$

so it cannot have full row rank. This is not merely an artifact of a careless choice of padding. Full row rank imposes genuine geometric restrictions.

Proposition 4.1 (A basic interlacing obstruction). *If $B = \begin{pmatrix} C \\ A \end{pmatrix}$ has full row rank, then A has full row rank and*

$$\kappa(B) \geq \kappa(A). \quad (19)$$

Proof. The matrix $BB^\top$ is positive definite and $AA^\top$ is a principal submatrix. Cauchy interlacing gives

$$\lambda_{\max}(BB^\top) \geq \lambda_{\max}(AA^\top), \quad \lambda_{\min}(BB^\top) \leq \lambda_{\min}(AA^\top).$$

Taking the ratio proves (19). $\square$

Thus the repaired smooth construction $g \equiv 0$ cannot yield $\kappa(B) = O(1)$ while $\kappa(A) \rightarrow \infty$ in the full-row-rank regime. For the nonsmooth hard family used above, one can say more.

For subspaces $U, V \subseteq \mathbb{R}^d$ with $U \cap V = \{0\}$, let

$$\theta(U, V) := \arccos \left(\max_{\substack{u \in U, v \in V \\ \|u\|=\|v\|=1}} |\langle u, v \rangle| \right) \in (0, \pi/2]$$

be their smallest principal angle.

Lemma 4.2 (Condition number versus principal angle). *If $B = \begin{pmatrix} C \\ A \end{pmatrix}$ has full row rank, then*

$$\kappa(B) \geq \cot \left(\frac{\theta(\text{row } C, \text{row } A)}{2} \right). \quad (20)$$

Proof. Let $u \in \text{row } C$ and $v \in \text{row } A$ be unit principal vectors with $\langle u, v \rangle = \cos \theta \geq 0$, where $\theta = \theta(\text{row } C, \text{row } A)$. Choose coefficient vectors p, q such that $C^\top p = u$ and $A^\top q = v$. The two coefficient vectors (p, q) and $(p, -q)$ have the same norm, while

$$B^\top(p, q) = u + v, \quad B^\top(p, -q) = u - v.$$

Therefore

$$\sigma_{\max}(B) \geq \frac{\|u + v\|}{\sqrt{\|p\|^2 + \|q\|^2}}, \quad \sigma_{\min}(B) \leq \frac{\|u - v\|}{\sqrt{\|p\|^2 + \|q\|^2}}.$$

Since $\|u + v\| = 2 \cos(\theta/2)$ and $\|u - v\| = 2 \sin(\theta/2)$, the ratio gives (20). $\square$

Lemma 4.3 (The row space required by an ℓ_1 seminorm). *Let $C_0 \in \mathbb{R}^{p \times d}$ have full row rank and let $g_0(x) = \lambda \|C_0 x\|_1$ with $\lambda > 0$. Suppose*

$$g_0(x) = \phi(Cx + c) \quad (21)$$

for some function ϕ , matrix C , and vector c . Then

$$\ker C \subseteq \ker C_0, \quad \text{equivalently} \quad \text{row } C_0 \subseteq \text{row } C. \quad (22)$$

Proof. If $h \in \ker C$, then (21) gives $g_0(h) = \phi(Ch + c) = \phi(c) = g_0(0) = 0$. Hence $\|C_0 h\|_1 = 0$, so $C_0 h = 0$. Taking orthogonal complements proves the row-space inclusion. $\square$

Proposition 4.4 (No full-row-rank reconditioning of the source hard family). *Consider the instances $\mathcal{P}_r$ in Proposition 3.1, with $m_1 = 2$. Let $g_r(x) = \phi(Cx + c)$ be any other valid representation of their same nonsmooth term. If $\begin{pmatrix} C \\ A_r \end{pmatrix}$ has full row rank, then*

$$\kappa \begin{pmatrix} C \\ A_r \end{pmatrix} \geq \kappa \begin{pmatrix} \bar{A}_r \\ A_r \end{pmatrix} \geq \frac{m}{4} = \frac{3r}{2}. \quad (23)$$

Proof. For $m_1 = 2$, the selected rows of the path-incidence matrix alternate: the rows of $\bar{A}_r$ are pairwise orthogonal, the rows of A_r are pairwise orthogonal, and every row in both blocks has the same norm. After a common normalization, both blocks therefore have orthonormal rows. If $\theta_r = \theta(\text{row } \bar{A}_r, \text{row } A_r)$, the block Gram matrix has the form

$$\begin{pmatrix} I & Q \\ Q^\top & I \end{pmatrix}, \quad \|Q\| = \cos \theta_r.$$

Its extreme eigenvalues are $1 \pm \cos \theta_r$, and hence

$$\kappa \begin{pmatrix} \bar{A}_r \\ A_r \end{pmatrix} = \sqrt{\frac{1 + \cos \theta_r}{1 - \cos \theta_r}} = \cot(\theta_r/2). \quad (24)$$

The nonsmooth term has the form $g_r(x) = \lambda_r \|\bar{A}_r x\|_1$ for some $\lambda_r > 0$. Lemma 4.3 therefore gives $\text{row } \bar{A}_r \subseteq \text{row } C$. Consequently,

$$\theta(\text{row } C, \text{row } A_r) \leq \theta_r.$$

Lemma 4.2, monotonicity of $\cot(\cdot/2)$, and (24) yield the first inequality in (23). The second is (12). $\square$

Remark 4.5 (Interpretation of Proposition 4.4). *The source hard family itself already satisfies the two contextual assumptions: its original outer function is Lipschitz and its original stacked matrix has full row rank. Proposition 4.4 shows that, for this family, the large stacked condition number is not removable by choosing a different full-row-rank representation. Thus the simple representation pathology is eliminated precisely in the regime highlighted by the authors. Establishing or refuting an Algorithm Class 1 upper bound in that regime requires a genuinely new algorithmic or lower-bound idea.*

5 Relation to subsequent near-optimal algorithms

A later paper by the same authors [2] proves a near-optimal upper bound under additional assumptions, but its oracle exposes the factorization: it returns, among other quantities, $\bar{A}x$, $\bar{A}^\top y$, and proximal maps of the outer function. This is not the exact Algorithm Class 1 model (5)–(6), whose nonsmooth access is only $\text{prox}_{\eta g}$. Accordingly, that result does not close the full-row-rank Algorithm Class 1 question considered here.

6 Corrections to the preliminary draft

The proof above replaces, rather than merely edits, the preliminary smooth-case argument. Three corrections are essential.

1. **Equality-constraint rescaling is not harmless.** Replacing (A, b) by (cA, cb) preserves the exact feasible set and $\kappa(A)$, but it changes the approximate feasibility requirement in (3) from $\|Ax + b\| \leq \varepsilon$ to $|c| \|Ax + b\| \leq \varepsilon$. It therefore cannot be used to normalize $\|A\|$ in an oracle-complexity statement at fixed ε . Lemma 2.1 instead chooses $\rho \geq \|C_\alpha\|$ and never changes the constraint.
2. **The exact algorithm class matters.** A list of available first-order quantities is not enough to invoke a lower bound. The present proof uses the precise spans (5)–(6) and imports the lower bound proved for exactly that class in [1].
3. **The lower-bound quantifiers are now correct.** For each candidate method and each r , the cited hard-family result applies to all Algorithm Class 1 methods on the specified instance. No claim of a single universal instance defeating every conceivable oracle model is needed, and no appeal to the broader wording “any matrix-vector method” is made.

7 Conclusion

The represented parameter $\kappa\left(\begin{pmatrix} \bar{A} \\ A \end{pmatrix}\right)$ is degenerate over unrestricted Problem Class 1 representations: ignored outer coordinates can drive it to one while leaving every piece of information available to Algorithm Class 1 unchanged. The authors’ own hard family then gives a rigorous contradiction to any uniform $\tilde{O}(\kappa(B)L_f\Delta F_0\varepsilon^{-2})$ upper bound. This is a negative solution to the literal open question.

The conclusion changes under the full-row-rank interpretation suggested by the surrounding text. Padding is then inadmissible, $\kappa(B) \geq \kappa(A)$, and the source hard family possesses an intrinsic principal-angle obstruction that prevents any full-row-rank re-representation from reducing its condition number. The contextual full-row-rank conjecture therefore remains unresolved; the present results constitute partial progress by isolating the exact formulation defect and proving that the obvious representation-based attack cannot cross the missing hypothesis.

References

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